Solar Si appendix

Sun · Si · downloadable appendix

Document	Solar Si appendix
Subject	Solar photospheric abundance appendix · Si
Author	Ryan Schmitt
Date	2026-10-28
Publisher	The Exoplanet Codex · exoplanetcodex.org
Version	feed v1.278
Source	science commit 27afcfcba7db
Web	https://exoplanetcodex.org/systems/sol/elements/si/

Solar silicon: 7.487 ± 0.086

Asplund, Amarsi & Grevesse 2021: 7.51 ± 0.03 · this measurement -0.023 dex

Our best HARPS product, chosen the same way as for iron: the Reference Grade analysis with the most complete treatment, then the most lines. HARPS is the instrument we use for the stars that follow, so the Sun is anchored on it. Products and engines are not averaged together.

VIS: 7.487 ± 0.086 · HARPS · VIS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)

Highlighted Products

VIS

7.489 ± 0.074

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

7.577 ± 0.127

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

NIR

7.522 ± 0.069

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

H

7.325 ± 0.104

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · CRIRES+ H · Reference Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	38 products -- Si I · Asplund et al. 2021 lines; Si I · Deshmukh et al. 2022 lines; Si I · Elgueta et al. 2026 lines; Si II · Asplund et al. 2021 lines

Grade	What it means	This element
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	24 products -- Si I · Codex-graded lines; Si II · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

VIS — HARPS
A(Si) dex; axis 7.13–7.85

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.535 ±0.173
Si II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		7.511 ±0.103
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.514 ±0.080
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.487 ±0.086
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.514 ±0.080
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.487 ±0.086

VIS — IAG FTS
A(Si) dex; axis 7.13–7.85

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.519 ±0.139
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.489 ±0.146
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.511 ±0.072
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.484 ±0.078
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.511 ±0.072
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.484 ±0.078

VIS — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.13–7.85

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.528 ±0.177
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.491 ±0.327
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.513 ±0.068
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.486 ±0.075
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.513 ±0.068
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.486 ±0.075

VIS — Kitt Peak (1984, molecfit)
A(Si) dex; axis 7.13–7.85

Si II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		7.518 ±0.155
Si II · Codex-graded lines · 1D LTE Codex Grade · 3 lines		7.488 ±0.303
Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		7.516 ±0.067
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		7.489 ±0.074
Si I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		7.516 ±0.067
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		7.489 ±0.074

VIS axis: 7.13 · 7.31 · 7.49 · 7.67 · 7.85 A(Si) dex

red-optical — IAG FTS
A(Si) dex; axis 7.41–7.75

Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.591 ±0.125
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.573 ±0.128

red-optical — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.41–7.75

Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.581 ±0.125

red-optical — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 7.41–7.75

Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.563 ±0.128

red-optical — Kitt Peak (1984, molecfit)
A(Si) dex; axis 7.41–7.75

Si I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		7.595 ±0.123
Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.577 ±0.127

red-optical axis: 7.41 · 7.49 · 7.58 · 7.66 · 7.75 A(Si) dex

NIR — CRIRES+ J
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 2 lines		7.513 ±0.086
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.487 ±0.092
Si I · Codex-graded lines · 1D LTE Codex Grade · 27 lines		7.527 ±0.103
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 27 lines		7.507 ±0.107

NIR — CRIRES+ Y
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 1 line		7.468 ±0.067
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.428 ±0.073
Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 2 lines		7.457 ±0.082
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		7.430 ±0.087
Si I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		7.582 ±0.113
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 4 lines		7.560 ±0.120

NIR — CRIRES+ Y (full arm)
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 1 line		7.468 ±0.067
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.428 ±0.073
Si I · Codex-graded lines · 1D LTE Codex Grade · 8 lines		7.551 ±0.084
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 8 lines		7.523 ±0.091

NIR — IAG FTS
A(Si) dex; axis 6.95–8.11

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 1 line		7.554 ±0.140
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.542 ±0.143
Si I · Codex-graded lines · 1D LTE Codex Grade · 13 lines		7.528 ±0.161
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 9 lines		7.497 ±0.147

NIR — Kitt Peak (Kurucz 2005)
A(Si) dex; axis 6.95–8.11

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 1 line		7.558 ±0.175
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		7.546 ±0.178
Si I · Codex-graded lines · 1D LTE Codex Grade · 13 lines		7.558 ±0.080
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 9 lines		7.547 ±0.106

NIR — Kitt Peak (1984, molecfit)
A(Si) dex; axis 6.95–8.11

Si I · Deshmukh et al. 2022 lines · 1D LTE Reference Grade · 3 lines		7.581 ±0.075
Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.552 ±0.083
Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 15 lines		7.536 ±0.063

NIR — Kitt Peak (1984, molecfit)
A(Si) dex; axis 6.95–8.11

Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 15 lines		7.522 ±0.069
Si I · Codex-graded lines · 1D LTE Codex Grade · 46 lines		7.536 ±0.516
Si I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 43 lines		7.529 ±0.553

NIR axis: 6.95 · 7.24 · 7.53 · 7.82 · 8.11 A(Si) dex

H — CRIRES+ H
A(Si) dex; axis 7.19–7.59

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 3 lines		7.363 ±0.101
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.325 ±0.104

H — CRIRES+ H (full arm, telluric-corrected)
A(Si) dex; axis 7.19–7.59

Si I · Elgueta et al. 2026 lines · 1D LTE Reference Grade · 3 lines		7.418 ±0.121
Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		7.380 ±0.122

H axis: 7.19 · 7.29 · 7.39 · 7.49 · 7.59 A(Si) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the total uncertainty -- the statistical and systematic terms combined in quadrature, as Asplund et al. 2021 and Lodders et al. 2025 quote it. The full error budget remains attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every silicon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the silicon abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Si)	Largest uncertainty terms (dex)
VIS	Si I · Codex-graded lines; Si II · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2–7	7.484–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03
VIS	Si I · Asplund et al. 2021 lines; Si II · Asplund et al. 2021 lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–7	7.484–7.489 (1D-NLTE)	line strengths (gf) 0.05, continuum placement 0.03
red-optical	Si I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2	7.563–7.577 (1D-NLTE)	line strengths (gf) 0.11, fit / line-to-line scatter 0.04
NIR	Si I · Codex-graded lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	4–46	7.497–7.560 (1D-NLTE)	telluric residual 0.55, line strengths (gf) 0.04
NIR	Si I · Deshmukh et al. 2022 lines	CRIRES+ J, CRIRES+ Y, CRIRES+ Y (full arm), Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	1–3	7.428–7.552 (1D-NLTE)	line strengths (gf) 0.06, fit / line-to-line scatter 0.04
NIR	Si I · Elgueta et al. 2026 lines	CRIRES+ Y, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–15	7.430–7.546 (1D-NLTE)	line strengths (gf) 0.04, fit window 0.03
H	Si I · Elgueta et al. 2026 lines	CRIRES+ H, CRIRES+ H (full arm, telluric-corrected)	1D-LTE, 1D-NLTE	3	7.325–7.380 (1D-NLTE)	continuum placement 0.06, line strengths (gf) 0.05

Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 59 lines.

Abundance outlier: disagrees with the other lines well beyond its error, and no solar analysis in the literature uses it.

Not measurable: the line falls in a data gap or a saturated telluric band.

Saturated: on the flat part of the curve of growth, so the line barely responds to the abundance (reduced width log W/λ above -4.9).

Species	Wavelength (Å, air)	Why it is excluded
Si II	3203.872	saturated (log W/λ = -4.18)
Si II	3210.028	saturated (log W/λ = -3.98)
Si II	3339.818	saturated (log W/λ = -4.04)
Si II	3856.017	saturated (log W/λ = -3.90)
Si II	3862.595	saturated (log W/λ = -4.15)
Si I	3905.523	saturated (log W/λ = -3.85)
Si I	4102.936	saturated (log W/λ = -4.10)
Si I	5948.541	saturated (log W/λ = -4.79)
Si I	7918.383	saturated (log W/λ = -4.88)

Species	Wavelength (Å, air)	Why it is excluded
Si I	7932.348	saturated (log W/λ = -4.81)
Si I	7944.001	saturated (log W/λ = -4.75)
Si I	8543.151	saturated (log W/λ = -4.23)
Si I	8648.465	saturated (log W/λ = -4.72)
Si I	8661.177	saturated (log W/λ = -4.27)
Si I	8728.010	saturated (log W/λ = -4.89)
Si I	8742.446	saturated (log W/λ = -4.90)
Si I	8752.007	saturated (log W/λ = -4.79)
Si I	8790.389	saturated (log W/λ = -4.82)
Si I	9021.590	saturated (log W/λ = -4.88)
Si I	9413.500	saturated (log W/λ = -4.56)
Si I	9887.047	abundance outlier
Si I	10313.197	abundance outlier
Si I	10371.263	saturated (log W/λ = -4.73)
Si I	10585.141	saturated (log W/λ = -4.46)
Si I	10603.425	saturated (log W/λ = -4.57)
Si I	10627.648	saturated (log W/λ = -4.85)
Si I	10660.973	saturated (log W/λ = -4.53)
Si I	10689.716	saturated (log W/λ = -4.70)
Si I	10694.251	saturated (log W/λ = -4.66)
Si I	10727.406	saturated (log W/λ = -4.62)
Si I	10749.378	saturated (log W/λ = -4.54)
Si I	10786.849	saturated (log W/λ = -4.59)
Si I	10827.088	saturated (log W/λ = -4.40)
Si I	10843.858	saturated (log W/λ = -4.73)
Si I	10868.790	saturated (log W/λ = -4.49)
Si I	10869.536	saturated (log W/λ = -4.34)
Si I	10885.333	saturated (log W/λ = -4.84)
Si I	10979.308	saturated (log W/λ = -4.62)
Si I	10982.058	saturated (log W/λ = -4.90)
Si I	11017.966	saturated (log W/λ = -4.68)
Si I	11187.588	saturated (log W/λ = -4.89)
Si I	11289.829	saturated (log W/λ = -4.86)
Si I	11611.060	saturated (log W/λ = -4.60)
Si I	11984.198	saturated (log W/λ = -4.46)
Si I	11991.568	saturated (log W/λ = -4.51)
Si I	12031.504	saturated (log W/λ = -4.38)
Si I	12103.534	saturated (log W/λ = -4.59)
Si I	12189.241	abundance outlier
Si I	12270.692	saturated (log W/λ = -4.62)
Si I	15077.332	saturated (log W/λ = -4.67)
Si I	15557.778	saturated (log W/λ = -4.88)
Si I	15888.409	saturated (log W/λ = -4.39)

Species	Wavelength (Å, air)	Why it is excluded
Si I	15960.063	saturated (log W/λ = -4.53)
Si I	16060.009	saturated (log W/λ = -4.64)
Si I	16163.691	saturated (log W/λ = -4.78)
Si I	16380.176	saturated (log W/λ = -4.72)
Si I	16381.534	saturated (log W/λ = -4.54)
Si I	16434.927	not measurable
Si I	17466.879	saturated (log W/λ = -4.45)

Solar silicon context

Silicon is traced by Si I lines from the blue to the H band and by the single Si II 6371 Å line. The optical lines are Asplund et al. 2021's set (Amarsi & Asplund 2017) with the gf the authors used; the infrared lines are the published sets of Deshmukh et al. 2022 and Elgueta et al. 2026 on Pehlivan Rhodin et al. 2024 oscillator strengths. Each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Si appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 27afcfcba7db6b4782ca756c1131fb9ddaa8dc3c
Feed Si.json v1.278 · updated 2026-10-17T23:20:32Z
Generated 2026-10-28T11:24:25-06:00 (source commit timestamp)
Feed SHA-256 7c1195f65f701ad6d0d6dbd97f55dee5cfdd7b07b6f7a1291b6ab2ce2619f708
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Si comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set SI_AGSS21 (the line selection). Sources: [3].

Atomic data and gf authorities. Si II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3], [4], [5].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [6].

Solar abundance and indicator analyses. Published line set SI_ELGUETA2026 (the line selection). Sources: [8].

Atomic data and gf authorities. Si I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [9].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Si_Amarsi2020_PySME.csv). Sources: [10].

Solar abundance and indicator analyses. Published line set SI_DESHMUKH2022 (the line selection). Sources: [11].

Product-to-source index

VIS · HARPS · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-18313ebf6e74

VIS · HARPS · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-a600a1f3ca00

VIS · IAG FTS · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-7b7e31f61f2b

VIS · IAG FTS · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-d538a2159904

VIS · Kitt Peak (Kurucz 2005) · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-0d051b970724

VIS · Kitt Peak (Kurucz 2005) · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-c60cd632806c

VIS · Kitt Peak (1984, molecfit) · Si II · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-45b0eb7e0d7e

VIS · Kitt Peak (1984, molecfit) · Si II · Codex-graded lines · 1D LTE · [1], [2], [7], [3], [4], [5], [6] Si-products-eb4fab3570b0

H · CRIRES+ H · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-5005aa52ea70

H · CRIRES+ H · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-c6dd0223e3f3

H · CRIRES+ H (full arm, telluric-corrected) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-e636e6942aaf

H · CRIRES+ H (full arm, telluric-corrected) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-50ccbe6c093c

NIR · CRIRES+ J · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-78ada3f9735b

NIR · CRIRES+ J · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-b8216b0c9359

NIR · CRIRES+ J · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-d56ffa9cfc5d

NIR · CRIRES+ J · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-6087fbe44c59

NIR · CRIRES+ Y · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-68a0da419260

NIR · CRIRES+ Y · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-fe06e8201b25

NIR · CRIRES+ Y · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-d01c0de482f8

NIR · CRIRES+ Y · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-bcdae8eb6758

NIR · CRIRES+ Y · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-02cc90577c3c

NIR · CRIRES+ Y · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-d9c04fe0bb0b

NIR · CRIRES+ Y (full arm) · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-115af02fe503

NIR · CRIRES+ Y (full arm) · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-e9b064f45645

NIR · CRIRES+ Y (full arm) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-f29c5165a798

NIR · CRIRES+ Y (full arm) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-ef230dcf5d1d

NIR · IAG FTS · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-dab7a7917e06

NIR · IAG FTS · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-319f483148d1

NIR · IAG FTS · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-aa27575bc65f

NIR · IAG FTS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-703e8de44fc8

NIR · Kitt Peak (Kurucz 2005) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-b1291f7f286b

NIR · Kitt Peak (Kurucz 2005) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-f403a4693354

NIR · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-09438da36aef

NIR · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-63a93f8943d7

NIR · Kitt Peak (1984, molecfit) · Si I · Deshmukh et al. 2022 lines · 1D LTE · [1], [2], [11], [9], [6] Si-products-1a21e353cd5f

NIR · Kitt Peak (1984, molecfit) · Si I · Deshmukh et al. 2022 lines · 1D NLTE (Amarsi) · [1], [2], [11], [9], [10] Si-products-939560fefceb

NIR · Kitt Peak (1984, molecfit) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-products-01a5009be9be

NIR · Kitt Peak (1984, molecfit) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-products-724d64a49bde

NIR · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D LTE · [1], [2], [9], [6] Si-products-a23c25d935d8

NIR · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10] Si-products-83f80d5ca2c1

VIS · HARPS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-b2f34d015eab

VIS · HARPS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-04b5965513ce

VIS · HARPS · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-93a85c979084

VIS · HARPS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-ea8e9f9add24

VIS · IAG FTS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-69325d3ca3fa

VIS · IAG FTS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-ae649028ea9a

VIS · IAG FTS · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-b8d5fe63d15d

VIS · IAG FTS · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-4ac612585049

VIS · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-71ea831123e9

VIS · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-fab5c8319865

VIS · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-47f5784280f5

VIS · Kitt Peak (Kurucz 2005) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-c16c0efa9866

VIS · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-e2f56f577f7c

VIS · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-a54926dacc0b

VIS · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-784a936bcab2

VIS · Kitt Peak (1984, molecfit) · Si I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-dcd2f1d1bede

red-optical · IAG FTS · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-e040cb5e6430

red-optical · IAG FTS · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-0b7df139857f

red-optical · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-c31c1e8e3ffd

red-optical · Kitt Peak (Kurucz 2005) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-8fce3970b2cc

red-optical · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4], [5], [6] Si-products-4eadb43adaf9

red-optical · Kitt Peak (1984, molecfit) · Si I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [4], [5], [10] Si-products-ae5c932e8b0b

NIR · CRIRES+ Y (full arm) · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-quarantine-7ebb8d8816da

NIR · CRIRES+ Y (full arm) · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-quarantine-c8be7c1b1687

NIR · CRIRES+ J · Si I · Elgueta et al. 2026 lines · 1D LTE · [1], [2], [8], [9], [6] Si-quarantine-6171b736d381

NIR · CRIRES+ J · Si I · Elgueta et al. 2026 lines · 1D NLTE (Amarsi) · [1], [2], [8], [9], [10] Si-quarantine-b905b0754458

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites. Space Science Reviews 221 DOI: 10.1007/s11214-025-01146-w. source record

[3] — Amarsi, A. M.; Asplund, M. (2017). The solar silicon abundance based on 3D non-LTE calculations. MNRAS 464, 264-273 DOI: 10.1093/mnras/stw2445. source record

[8] — Elgueta, S.; Jofre, P.; Aguilera-Gomez, C.; Slumstrup, D.; Rojas Arriagada, A.; Heiter, U.; Casamiquela, L.; Zoccali, M.; Worley, C.; Soubiran, C. (2026). Gaia FGK benchmark stars: Selecting infrared lines for abundance determination. A&A 710, A111 DOI: 10.1051/0004-6361/202659148. source record

[11] — Deshmukh, S. A.; Ludwig, H.-G.; Kucinskas, A.; et al. (2022). The solar photospheric silicon abundance according to CO5BOLD: Investigating line broadening, magnetic fields, and model effects. A&A 668, A48 DOI: 10.1051/0004-6361/202142072. source record

Atomic data and gf authorities

[4] — Garz, T. (1973). Absolute oscillator strengths of Si I lines between 2500 A and 8000 A. A&A 26, 471-477 source record

[5] — O'Brian, T. R.; Wickliffe, M. E.; Lawler, J. E.; Whaling, W.; Brault, J. W. (1991). Lifetimes, transition probabilities, and level energies in Fe I. JOSA B 8, 1185 DOI: 10.1364/JOSAB.8.001185. source record

[7] — Kelleher, D. E.; Podobedova, L. I. (2008). Atomic Transition Probabilities of Silicon. A Critical Compilation. J. Phys. Chem. Ref. Data 37, 1285-1501 DOI: 10.1063/1.2734566. source record

[9] — Pehlivan Rhodin, A.; Hartman, H.; Nilsson, H.; Joensson, P. (2024). Accurate and experimentally validated transition data for Si I and Si II. A&A 682, A184 DOI: 10.1051/0004-6361/202245686. source record

Model and grid sources

[6] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

[10] — Amarsi, A. M.; Lind, K.; Osorio, Y.; Nordlander, T.; Bergemann, M.; et al. (2020). The GALAH Survey: non-LTE departure coefficients for large spectroscopic surveys. A&A 642, A62 DOI: 10.1051/0004-6361/202038650. source record